J Geogr Syst
DOI 10.1007/s10109-015-0213-7

ORIGINAL ARTICLE

Random effects specifications in eigenvector spatial filtering: a simulation study

Daisuke Murakami[1] · Daniel A. Griffith[2]

Received: 13 September 2013 / Accepted: 27 April 2015
© Springer-Verlag Berlin Heidelberg 2015

Abstract Eigenvector spatial filtering (ESF) is becoming a popular way to address spatial dependence. Recently, a random effects specification of ESF (RE-ESF) is receiving considerable attention because of its usefulness for spatial dependence analysis considering spatial confounding. The objective of this study was to analyze theoretical properties of RE-ESF and extend it to overcome some of its disadvantages. We first compare the properties of RE-ESF and ESF with geostatistical and spatial econometric models. There, we suggest two major disadvantages of RE-ESF: it is specific to its selected spatial connectivity structure, and while the current form of RE-ESF eliminates the spatial dependence component confounding with explanatory variables to stabilize the parameter estimation, the elimination can yield biased estimates. RE-ESF is extended to cope with these two problems. A computationally efficient residual maximum likelihood estimation is developed for the extended model. Effectiveness of the extended RE-ESF is examined by a comparative Monte Carlo simulation. The main findings of this simulation are as follows: Our extension successfully reduces errors in parameter estimates; in many cases, parameter estimates of our RE-ESF are more accurate than other ESF models; the elimination of the spatial component confounding with explanatory variables results in biased parameter estimates; efficiency of an accuracy maximization-based conventional ESF is comparable to RE-ESF in many cases.

✉ Daisuke Murakami
murakami.daisuke@nies.go.jp

Daniel A. Griffith
dagriffith@utdallas.edu

[1] Center for Global Environmental Research, National Institute for Environmental Studies, 16-2 Onogawa, Tsukuba-shi, Ibaraki 305-8506, Japan

[2] School of Economic, Political and Policy Sciences, The University of Texas at Dallas, Richardson, Dallas, TX 75083, USA

Published online: 20 May 2015

Keywords Eigenvector spatial filtering · Mixed effects model · Geostatistics · Spatial econometrics · Spatial confounding

JEL Classification C15 · C21

1 Introduction

Along with the recent development of geographic information systems (GIS), statistical methods for spatial and spatiotemporal data have been discussed extensively in geostatistics (e.g., Cressie 1993; Cressie and Wikle 2011) and spatial econometrics (e.g., LeSage and Pace 2009). Geostatistical and spatial econometric methods are useful because they consider properties of spatial data such as spatial dependence and spatial heterogeneity. Spatial filtering (e.g., Griffith 2010; Griffith and Paelinck 2011; Griffith and Chun 2013), which relates to both geostatistics and spatial econometrics, is an approach that captures spatial dependence using map pattern variables obtained from spatial connectivity information (Getis and Griffith 2002). The G statistic (Getis and Ord 1992)-based filtering method of Getis (1990) and the Moran coefficient (Moran 1950)-based method called eigenvector spatial filtering (ESF; Griffith 1996) are representative methods. Applications of ESF are increasing because of its practicality, expandability, and other appealing factors. For instance, Chun (2008), Fischer and Griffith (2008), Griffith (2009, 2011), Chun and Griffith (2011), Scherngell and Lata (2013), and Griffith and Fischer (2013) utilize ESF for spatial interaction analyses; Grimpe and Patuelli (2009), Patuelli et al. (2011), and Cuaresma and Feldkircher (2013) employ it for economic analyses; Griffith and Peres-Neto (2006) and Jacob et al. (2008) use it for ecological analyses; Helbich and Arsanjani (2014) and Chun (2014) use it for criminal analyses; Wang et al. (2013) analyze land value and land use applying it; and Moniruzzaman and Paez (2012) use it for urban design analysis. Thus, ESF has become a popular method to address spatial dependence (Pace et al. 2013).

One of the advantages of ESF is simplicity (Griffith 2003). The basic model of ESF is identical to the standard linear and generalized linear regression models, and therefore, it is easily implemented and extended, for example, by combining it with standard non-spatial models (e.g., Poisson and logistic regression; see, e.g., Griffith 2002, 2004a). Another advantage of ESF is its effectiveness in capturing spatial dependence. Tiefelsdorf and Griffith (2007) show that ESF effectively removes spatial dependence from residuals, and Griffith (2006) demonstrates its usefulness for extracting negative spatial dependence hidden by predominant positive spatial dependence. Thayn and Simanis (2013) show that ESF reduces spatial misspecification errors, increases the strength of a model fit, frequently increases the normality of model residuals, and can increase the homoscedasticity of model residuals. Franzese and Hays (2014) show the effectiveness of ESF in distinguishing a common exposure effect, which is a source of spatial dependence, and a spatial spillover effect, which is another source of spatial dependence. Some important

statistical properties of ESF are discussed in Chun and Griffith (2013, 2014) as well as in Pace et al. (2013).

Hughes and Haran (2013) discuss a random effects specification of ESF (RE-ESF) and demonstrate the usefulness for spatial dependence analysis while mitigating the variance inflation due to the confounding factor between explanatory variables and the latent spatial process. The confounding factor is called spatial confounder in the statistical literature (e.g., Paciorek 2010; Hodges and Reich 2010). However, discussion of RE-ESF is limited, and its theoretical properties are not necessarily thoroughly understood.

The objective of this study was to analyze advantages and disadvantages of RE-ESF, and extend it to overcome some of the disadvantages. This paper's sections are organized as follows. Section 2 briefly explains conventional ESF and RE-ESF. Section 3 analyzes theoretical properties of ESF and RE-ESF models by comparing them with models in geostatistics and spatial econometrics. Therein, we point to two disadvantages of RE-ESF: it is sensitive to the chosen spatial connectivity structure, and while the current form of RE-ESF eliminates spatial confounder, the elimination can yield biased parameter estimates. Section 4 extends RE-ESF to cope with these problems, and Sect. 5 compares ESF models, including our extended RE-ESF, by a Monte Carlo simulation. Finally, Sect. 6 concludes our discussion.

2 Eigenvector spatial filtering: ESF

2.1 The Moran coefficient: MC

The Moran coefficient (MC), given by the following equation, is a spatial dependence diagnostic statistic:

$$\mathrm{MC} = \frac{N}{\mathbf{1}'\mathbf{C}\mathbf{1}} \frac{\mathbf{z}'\mathbf{M}\mathbf{C}\mathbf{M}\mathbf{z}}{\mathbf{z}'\mathbf{M}\mathbf{z}}, \tag{1}$$

where "′" represents matrix transpose, $\mathbf{1}$ is a $N \times 1$ vector of ones, $\mathbf{z}$ is a $N \times 1$ vector of variable values, $\mathbf{C}$ is a $N \times N$ symmetric connectivity matrix whose diagonal elements are zero, and $\mathbf{M}$ is a $N \times N$ projection matrix. Two types of projection matrix $\mathbf{M}$, $\mathbf{I}–\mathbf{1}\mathbf{1}'/N$ and $\mathbf{I}–\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, have been applied, where $\mathbf{I}$ is an $N \times N$ identity matrix, and $\mathbf{X}$ is a $N \times K$ matrix of K explanatory variables (e.g., Anselin and Rey 1991). Matrix $\mathbf{I}–\mathbf{1}\mathbf{1}'/N$ is used if $\mathbf{z}$ is adjusted for a constant mean, whereas $\mathbf{I}–\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is used if $\mathbf{z}$ is a residual vector of a linear regression model (e.g., adjusted for a variable mean). MC, which evaluates spatial dependence by measuring covariation among sample values, is positive if sample values in $\mathbf{z}$ display positive spatial dependence and negative if the sample values display negative spatial dependence.[1]

[1] The expectation of MC, $E[\mathrm{MC}]$, equals $-\frac{1}{N-1}$ when $\mathbf{M} = \mathbf{I}–\mathbf{1}\mathbf{1}'/N$, whereas it equals $-\frac{N}{\mathbf{1}'\mathbf{C}\mathbf{1}} \frac{tr[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{C}\mathbf{X}]}{N-K-1}$ when $\mathbf{M} = \mathbf{I}–\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$. $E[\mathrm{MC}] < \mathrm{MC}$, $\mathrm{MC} < E[\mathrm{MC}]$, and $\mathrm{MC} = E[\mathrm{MC}]$ imply positive, negative, and no spatial dependence, respectively.

2.2 Standard ESF

ESF models spatial dependence using the eigenvectors of **MCM** in Eq. (1). The eigenvector corresponding to the first eigenvalue, $\mathbf{e}_1$, is the set of real values that has the largest positive MC (depicting the maximum positive spatial dependence) achievable by any set of real numbers for the spatial arrangement defined by **C**. The second eigenvector, $\mathbf{e}_2$, is the set of real values that has the largest positive MC that is uncorrelated with and orthogonal to $\mathbf{e}_1$, and $\mathbf{e}_N$ is the set of numerical values that has the largest negative MC (depicting the maximum negative spatial dependence) achievable that is uncorrelated with and orthogonal to $\mathbf{e}_1, \ldots, \mathbf{e}_{N-1}$ (Griffith 2003). The set of eigenvectors of **MCM**, $\mathbf{E}_{\text{full}} = \{\mathbf{e}_1, \ldots, \mathbf{e}_N\}$, furnishes all possible distinct map pattern descriptions of latent spatial dependence, with each level being indexed by an MC that is proportional to its corresponding eigenvalue.

The basic linear model of ESF is

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{E}\gamma + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I}), \tag{2}$$

where $\mathbf{y}$ is a $N \times 1$ vector of response variable values; $\mathbf{E}$ is a $N \times L$ matrix composed of a subset of L eigenvectors ($L < N$) from $\mathbf{E}_{\text{full}}$; $\boldsymbol{\varepsilon}$ is a $N \times 1$ vector of disturbances; $\boldsymbol{\beta}$ and $\boldsymbol{\gamma} = [\gamma_1, \ldots, \gamma_l, \ldots, \gamma_L]'$ are parameter vectors whose sizes are $K \times 1$ and $L \times 1$, respectively; σ^2 is a variance parameter; and $\mathbf{0}$ is a $N \times 1$ vector of zeros. Equation (2) is a semiparametric model. For $\mathbf{M} = \mathbf{I}–\mathbf{11}'/N$, Eq. (2) is an approximation of a standard spatial econometric model called the spatial lag model, whereas, when $\mathbf{M} = \mathbf{I}–\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, Eq. (2) is an approximation of the spatial error model, another spatial econometric model (Tiefelsdorf and Griffith 2007). When $\mathbf{M} = \mathbf{I}–\mathbf{X}(\mathbf{X}'\ \mathbf{X})^{-1}\mathbf{X}'$, the eigenvectors are mutually uncorrelated as well as uncorrelated with **X**.

The L eigenvectors in **E** are selected as follows: (1) Eigenvectors representing inconsequential levels of spatial dependence are removed, and (2) significant eigenvectors are chosen using a stepwise selection method. Generally, step (1) is conducted by removing eigenvectors whose eigenvalues are small or of the wrong nature, whereas step (2) is done by maximizing model accuracy or minimizing residual spatial dependence. The adjusted R-square is a commonly used objective function for the former, and residual MC is the objective function for the latter. Because Eq. (2) is identical to the standard linear regression model, step (2) can be conducted by using ordinary least squares (OLS) estimation-based stepwise methods. OLS estimators of $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ are given as

$$\begin{bmatrix} \hat{\beta} \\ \hat{\gamma} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{E} \\ \mathbf{E}'\mathbf{X} & \mathbf{I} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{X}'\mathbf{y} \\ \mathbf{E}'\mathbf{y} \end{bmatrix}. \tag{3}$$

Such stepwise techniques also are available for generalized linear models.

2.3 RE-ESF

The linear model of RE-ESF (Hughes and Haran 2013), which is based on a model with an intrinsic conditional autoregressive prior, is formulated as follows:

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{E}\gamma + \boldsymbol{\varepsilon}, \quad \boldsymbol{\gamma} \sim N(\mathbf{0}, \sigma_\gamma^2 \boldsymbol{\Lambda}), \quad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \boldsymbol{\sigma}^2 \mathbf{I}) \tag{4}$$

where σ_γ^2 is a variance parameter, and $\boldsymbol{\Lambda}$ is a $L \times L$ diagonal matrix whose elements are the first L-eigenvalues of **MCM**, $\lambda_1, \ldots, \lambda_L$, corresponding to the L eigenvectors in **E**. Equation (4) is identical to the standard linear mixed model (LMM). Hughes and Haran (2013) introduce eigenvectors into **E** whose eigenvalues are greater than zero. They estimate parameters with Markov Chain Monte Carlo (MCMC) techniques.

The best linear unbiased estimators (BLUEs) of $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ are obtained as (see Henderson 1975)[2]

$$\begin{bmatrix} \hat{\beta} \\ \hat{\gamma} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{E} \\ \mathbf{E}'\mathbf{X} & \mathbf{I} + \dfrac{\sigma^2}{\sigma_\gamma^2}\boldsymbol{\Lambda}^{-1} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{X}'\mathbf{y} \\ \mathbf{E}'\mathbf{y} \end{bmatrix}, \tag{5}$$

where $\boldsymbol{\Lambda}^{-1}$ is a $L \times L$ diagonal matrix whose elements are $1/\lambda_1, \ldots, 1/\lambda_L$. BLUEs of the standard ESF, Eq. (3), converge to Eq. (5) as σ_γ^2 goes to $+\infty$. In other words, the conventional ESF is a limiting case of RE-ESF. In Eq. (5), the l-th diagonal ($l = 1, \ldots, L$) of $(\sigma^2/\sigma_\gamma^2)\boldsymbol{\Lambda}^{-1}$, which is given by $(\sigma^2/\sigma_\gamma^2)\ \lambda_l^{-1}$, shrinks γ_l toward zero. The shrinkage is significant for eigenvectors corresponding to smaller eigenvalues, λ_l. Such shrinkage, which is useful for model estimation while holding constant the degrees of freedom (Hodges 2013), allows estimation of the RE-ESF model without applying a stepwise eigenvector selection.

Variances of the estimated $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ are given as follows

$$\mathrm{Var}\begin{bmatrix} \hat{\beta} \\ \hat{\gamma} \end{bmatrix} = \sigma^2 \begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{E} \\ \mathbf{E}'\mathbf{X} & \mathbf{I} + \dfrac{\sigma^2}{\sigma_\gamma^2}\boldsymbol{\Lambda}^{-1} \end{bmatrix}^{-1}. \tag{6}$$

3 Properties of the standard ESF and RE-ESF

3.1 Relationship with geostatistical models

Consider a standard geostatistical model with known mean,[3] which is formulated as

$$\mathbf{y} = \mathbf{X}\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}} + \boldsymbol{\omega}, \quad \boldsymbol{\omega} \sim N(\mathbf{0}, \sigma^2\mathbf{I} + \sigma_\gamma^2\mathbf{C}), \tag{7}$$

where $\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}} = \mathbf{X}(\mathbf{X}'\ \mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ are the OLS estimates. $\boldsymbol{\omega}$ in Eq. (7) describes a stationary Gaussian process (Cressie 1993). **C** in this model is a $N \times N$ covariance matrix whose (i, j)th element is given by a function of distance, such as the exponential function, $\exp(-d(s_i,s_j)/h)$, where $d(s_i, s_j)$ is the Euclidean distance between sample sites s_i and s_j, and h is a range parameter. Following ESF, diagonal elements

[2] $\hat{\beta}$ in Eq. (5) is identical to the generalized least squared estimator (Henderson 1975).

[3] Models with known mean structure are widely applied in geostatistics (e.g., Cressie and Wikle 2011).

of **C** are assumed to be zero.[4] A variant of ESF, which is called Moran's eigenvector maps (MEMs), uses the Euclidean distance-based **C** to extract spatial eigenvectors. MEMs have been applied for ecological data analysis, including spatial pattern detection and variation partitioning (e.g., Borcard and Legendre 2002; Peres-Neto et al. 2006; Blancheta et al. 2008; Legendre and Legendre 2012). Just like the standard ESF, which usually assumes a binary connectivity-based **C** matrix, MEMs assume $\mathbf{E}\boldsymbol{\gamma}$ as a fixed effects term and estimate $\boldsymbol{\gamma}$ via OLS.

Equation (7) can be expanded using $\mathbf{M} = \mathbf{I}\text{–}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ as

$$\boldsymbol{\omega} = \mathbf{M}\mathbf{y} \sim N(\mathbf{0}, \sigma^2\mathbf{M} + \sigma_\gamma^2\mathbf{MCM}). \tag{8}$$

Eigenvectors of **MCM** and $\sigma^2\mathbf{M} + \sigma_\gamma^2$ **MCM** are identical (Griffith 2003). Furthermore, eigenvalues of these matrices have the following relationship

$$\begin{aligned}\boldsymbol{\Lambda}_{\text{full}(+)} &= \sigma^2\mathbf{E}'_{\text{full}}\mathbf{M}\mathbf{E}_{\text{full}} + \sigma_\gamma^2\mathbf{E}'_{\text{full}}\mathbf{MCME}_{\text{full}},\\ &= \sigma^2\begin{pmatrix}\mathbf{E}'_{L^*}\\ \mathbf{E}'_{N-L^*}\end{pmatrix}\mathbf{M}(\,\mathbf{E}_{L^*} \quad \mathbf{E}_{N-L^*}\,) + \sigma_\gamma^2\boldsymbol{\Lambda}_{\text{full}},\\ &= \sigma^2\begin{pmatrix}\mathbf{I}_{L^*} & \mathbf{0}'\\ \mathbf{0} & \mathbf{0}_{N-L^*}\end{pmatrix} + \sigma_\gamma^2\boldsymbol{\Lambda}_{\text{full}},\end{aligned} \tag{9}$$

where $\boldsymbol{\Lambda}_{\text{full}(+)}$ is a diagonal matrix of eigenvalues of $\sigma^2\mathbf{M} + \sigma_\gamma^2\mathbf{MCM}$, and $\boldsymbol{\Lambda}_{\text{full}}$ is the same matrix of **MCM**. $\mathbf{E}_{\text{full}} = [\mathbf{E}_{L*}, \mathbf{E}_{N-L*}]$, where $\mathbf{E}_{L*}$ is the subset of $\mathbf{E}_{\text{full}}$ composed of L^* eigenvector whose eigenvalues are not zero ($L < L^*$), $\mathbf{E}_{N-L*}$ is the subset composed of the other eigenvectors, $\mathbf{I}_{L*}$ is an identity matrix, $\mathbf{0}$ and $\mathbf{0}_{N-L*}$ are matrixes of zeros. Equation (8) can be expanded by substituting Eq. (9) as

$$\boldsymbol{\omega} \sim N\left[\mathbf{0}, \mathbf{E}_{\text{full}}\left[\sigma^2\begin{pmatrix}\mathbf{I}_{L^*} & \mathbf{0}'\\ \mathbf{0} & \mathbf{0}_{N-L^*}\end{pmatrix} + \sigma_\gamma^2\boldsymbol{\Lambda}\right]\mathbf{E}'_{\text{full}}\right]. \tag{10}$$

Further, Eq. (10) can be approximated, by dropping $N - L$ non-principal eigenvectors, as

$$\boldsymbol{\omega} \sim N(\mathbf{0}, \sigma^2\mathbf{I} + \sigma_\gamma^2\mathbf{E}\boldsymbol{\Lambda}\mathbf{E}'). \tag{11}$$

Equation (11) may be rewritten as

$$\boldsymbol{\omega} = \mathbf{E}\gamma + \boldsymbol{\varepsilon}, \quad \boldsymbol{\gamma} \sim N(\mathbf{0}, \sigma_\gamma^2\boldsymbol{\Lambda}), \quad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I}). \tag{12}$$

By substituting Eq. (12) into the standard geostatistical model, Eq. (7), the model yields the RE-ESF model, Eq. (4). After all, RE-ESF is a low-rank approximation of the geostatistical model,[5] whereas the standard ESF is the limiting case, where σ_γ^2 goes to $+\infty$ (see Sect. 2.3).

[4] Although the covariance matrix of $\boldsymbol{\omega}$ usually is defined by $\sigma_{\text{NS}}^2\mathbf{I} + \sigma_\gamma^2(\mathbf{I} + \mathbf{C})$, where σ_{NS}^2 is a variance parameter, because it can be expanded as follows: $\sigma_{\text{NS}}^2\mathbf{I} + \sigma_\gamma^2(\mathbf{I} + \mathbf{C}) = (\sigma_{\text{NS}}^2 + \sigma^2\gamma)\mathbf{I} + \sigma_\gamma^2\mathbf{C} = \sigma^2\mathbf{I} + \sigma_\gamma^2\mathbf{C}$, the assumption of zero diagonal elements is consistent with the usual assumption.

[5] When $\mathbf{M} = \mathbf{I}\text{–}\mathbf{11}'/N$, RE-ESF approximates the geostatistical model with known constant mean.

This relationship justifies the use of a distance-based $\mathbf{C}$ following geostatistics, whereas ESF studies typically apply a binary connectivity-based $\mathbf{C}$. The relationship also highlights that ESF models must specify $\mathbf{C}$ a priori, whereas geostatistical approaches estimate $\mathbf{C}$ by parameterizing its elements (e.g., using $\exp(-d(s_i,s_j)/h)$). The next subsection discusses this problem from a spatial econometric perspective, and Sect. 4 how to cope with this problem.

3.2 Relationships with spatial econometric models

Spatial econometrics is another field concerned with the parameter estimation problem in the presence of spatial dependence. Spatial econometrics typically models spatial dependence by multiplying a variable vector $\mathbf{z}$ by $(\mathbf{I} - \rho\mathbf{W})^{-1} = \mathbf{I} + \rho\mathbf{W} + \rho^2\mathbf{W}^2\ldots$, which is a matrix describing spatial interactions, as $(\mathbf{I} - \rho\mathbf{W})^{-1}\mathbf{z}$, where ρ is a spatial autocorrelation parameter, and $\mathbf{W}$ $(= \mathbf{E}_\mathrm{W}\mathbf{\Lambda}_\mathrm{W}\mathbf{E}_\mathrm{W}')$[6] is a row-standardized spatial connectivity matrix. $(\mathbf{I} - \rho\mathbf{W})^{-1}\mathbf{z}$ can be expanded to $\mathbf{E}_\mathrm{W}\boldsymbol{\gamma}_\mathrm{W}$, which is similar to $\mathbf{E}\boldsymbol{\gamma}$ in ESF models, where $\boldsymbol{\gamma}_\mathrm{W} = (\mathbf{I} - \rho\mathbf{\Lambda}_\mathrm{W})^{-1}\mathbf{E}_\mathrm{W}'\mathbf{z}$ is a vector of coefficients of eigenvectors. Because $\boldsymbol{\gamma}_\mathrm{W}$ depends not only on data fitting, but also on the eigenvalues of $\mathbf{W}$, the choice of $\mathbf{W}$ must have a considerable impact on the resulting $\mathbf{E}_\mathrm{W}\boldsymbol{\gamma}_\mathrm{W}$. In fact, the sensitivity of spatial econometric models to the selection of $\mathbf{W}$ has been demonstrated in many studies (e.g., Arbia and Fingleton 2008; Elhorst 2010). LeSage and Pace (2014), however, illustrate that well-specified spatial econometric models are less sensitive to the selection of $\mathbf{W}$. But they do not discuss ESF models. Besides, they do not consider the influence of the confounding factor between explanatory variables and the spatial dependence component (i.e., spatial confounder: see Sect. 1).

For the ESF approach, it is also important to avoid misspecification of $\mathbf{C}$. Fortunately, $\boldsymbol{\gamma}$ in the standard ESF specification is estimated independent of the eigenvalues of $\mathbf{MCM}$, and the resulting $\mathbf{E}\boldsymbol{\gamma}$ is less sensitive to the choice of $\mathbf{C}$. In contrast, the estimate of $\boldsymbol{\gamma}$ in RE-ESF depends on the eigenvalues (see Eq. (5)); hence, the resulting $\mathbf{E}\boldsymbol{\gamma}$ would be more sensitive to the choice of $\mathbf{C}$.

Specification of $\mathbf{M}$ is another issue. $\mathbf{M} = \mathbf{I}–\mathbf{X}(\mathbf{X}'\ \mathbf{X})^{-1}\mathbf{X}'$ specifies $\mathbf{E}\boldsymbol{\gamma}$ to be orthogonal and uncorrelated with $\mathbf{X}$, whereas $\mathbf{M} = \mathbf{I}–\mathbf{1}\mathbf{1}'/N$ specifies $\mathbf{E}\boldsymbol{\gamma}$ such that it is not necessarily orthogonal and uncorrelated with $\mathbf{X}$. Hughes and Haran (2013) interpret the correlation between $\mathbf{E}\boldsymbol{\gamma}$ and $\mathbf{X}$ as a spatial confounder that inflates the uncertainty of the estimate of $\boldsymbol{\beta}$ and recommend using $\mathbf{M} = \mathbf{I}– \mathbf{X}(\mathbf{X}'\ \mathbf{X})^{-1}\mathbf{X}'$.

However, the use of $\mathbf{M} = \mathbf{I}–\ \mathbf{X}(\mathbf{X}'\ \mathbf{X})^{-1}\mathbf{X}'$ can artificially increase the significance of $\hat{\beta}$. To see this, consider the variance of $[\hat{\beta},\hat{\gamma}]'$, which is formulated by Eq. (6). When $\mathbf{M} = \mathbf{I}– \mathbf{X}(\mathbf{X}'\ \mathbf{X})^{-1}\mathbf{X}'$, Eq. (6) is expanded using $\mathbf{X}'\ \mathbf{E} = \mathbf{0}$, which $\mathbf{M} = \mathbf{I}– \mathbf{X}(\mathbf{X}'\ \mathbf{X})^{-1}\mathbf{X}'$ assumes, as follows:

$$\mathrm{Var}\begin{bmatrix}\hat{\boldsymbol{\beta}}\\ \hat{\boldsymbol{\gamma}}\end{bmatrix} = \sigma^2\begin{bmatrix}\mathbf{X}'\mathbf{X} & \mathbf{0}\\ \mathbf{0}' & \mathbf{I} + \dfrac{\sigma^2}{\sigma_{\hat{\gamma}}^2}\mathbf{\Lambda}^{-1}\end{bmatrix}^{-1}. \tag{13}$$

[6] The asymmetric matrix $\mathbf{W}$ has real eigenvalues and eigenvectors (see Griffith 2000).

Equation (13) implies $\mathrm{Var}[\hat{\beta}] = \sigma^2(\mathbf{X}'\ \mathbf{X})^{-1}$. The spatial effects term, $\mathbf{E}\boldsymbol{\gamma}$, decreases the residual variance, σ^2, and $\mathrm{Var}[\hat{\beta}] = \sigma^2(\mathbf{X}'\ \mathbf{X})^{-1}$ as well. As a result, $\mathbf{E}\boldsymbol{\gamma}$ inflates the significance of $\hat{\beta}$. However, the inflation is due to the assumption of $\mathbf{X}'\ \mathbf{E} = \mathbf{0}$ that is unrealistic in most real-world cases, in which the spatial dependence component can correlate with explanatory variables. In other words, the inflation might be an artifact introduced by the unrealistic assumption $\mathbf{X}'\ \mathbf{E} = \mathbf{0}$.

Spatial econometric studies (e.g., Pace and LeSage 2010) demonstrate the importance of considering spatial components correlated with $\mathbf{X}$; the consideration helps reducing omitted variables bias, which arises when some variables omitted from a model specification are correlated with $\mathbf{X}$. From a spatial econometric perspective, use of $\mathbf{M} = \mathbf{I}–\mathbf{11}'/N$, which does not eliminate spatial confounding, might be preferable. When $\mathbf{M} = \mathbf{I}–\mathbf{11}'/N$, $\mathbf{E}\boldsymbol{\gamma}$ results in a fixed or random intercept term capturing spatial heterogeneity.

In summary, while RE-ESF is theoretically consistent with geostatistics, some additional discussions of how to specify $\mathbf{C}$ and $\mathbf{M}$ are required.

4 An extension of RE-ESF

4.1 Model

This section discusses how to specify a spatial connectivity structure within a RE-ESF framework. In the spatial econometric literature, multiple spatial weight matrices often have been considered to specify $\mathbf{W}$ flexibly (e.g., LeSage and Parent 2007; Seya et al. 2014a). However, in an ESF framework, consideration of multiple $\mathbf{C}$ matrices requires multiple eigendecompositions, which increases its computational cost. Besides, use of eigenvectors of multiple $\mathbf{C}$ matrices decreases the degrees of freedom. Thus, we do not consider using multiple $\mathbf{C}$.

Instead, following some geostatistical studies (e.g., Pintore and Holmes 2004; Haskard and Lark 2009), we replace eigenvalues, λ_l, with $\lambda_l(\alpha) = \frac{\sum_l \lambda_l}{\sum_l \lambda_l^\alpha}\lambda_l^\alpha$, where α is an unknown parameter. α determines the spatial scale of spatial dependence; a large α describes global-scale spatial variations by assigning dominant powers for principal eigenvectors, whereas a small α describes local-scale spatial variations by assigning more powers for non-principal eigenvectors (see Fig. 1). The scaling term, $\Sigma_l\lambda_l/\Sigma_l\lambda_l^\alpha$, is introduced to increase the identifiability of α and σ_γ^2; it makes the total variation explained by $\lambda_l(\alpha)$, $\boldsymbol{\gamma}_0 \sim N(\mathbf{0}, \boldsymbol{\Lambda})$, constant regardless of α. As a result, the total variance of $\boldsymbol{\gamma} = \boldsymbol{\sigma}_\gamma\boldsymbol{\gamma}_0 \sim N(0, \sigma_\gamma^2\ \Lambda)$ depends solely on σ_γ^2, whereas the scale of spatial dependence depends only on α. As mentioned in Haskard and Lark (2009), the estimation of α is identical to the estimation of the range parameter in geostatistical models. Introduction of α would be useful to mitigate the sensitivity of RE-ESF to a chosen $\mathbf{C}$.

We propose an extended RE-ESF, which is defined as

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{E}\gamma + \boldsymbol{\varepsilon}, \quad \boldsymbol{\gamma} \sim N(\mathbf{0}, \sigma_\gamma^2\boldsymbol{\Lambda}(\alpha)), \quad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I}), \tag{14}$$

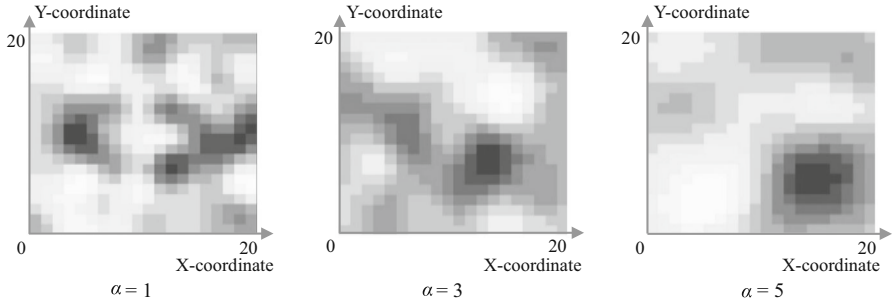

Fig. 1 Examples of $\mathbf{E}\boldsymbol{\gamma}$ generated in 20-by-20 grids: $\mathbf{E}\boldsymbol{\gamma}$ in the *left* ($\alpha = 1$), the *middle* ($\alpha = 3$), and the *right* ($\alpha = 5$) part of the figure describes local-, medium-, and global-scale spatial-dependent variations, respectively

where $\boldsymbol{\Lambda}(\alpha)$ is a $L \times L$ diagonal matrix whose l-th element is $\lambda_l(\alpha)$. Considering the consistency with geostatistics discussed in §3.1, elements in $\mathbf{C}$ are given by an exponential function, $\exp(-d(s_i,s_j)/h)$, and following some studies of distance-based spatial filtering (e.g., Griffith and Peres-Neto 2006; Dray et al. 2006; Griffith 2010), h in the function is the longest distance in the minimum spanning tree covering the N observation sites. Equation (14) approximates the standard geostatistical model with the exponential function that is well approximated by the conditional autoregressive model with a binary connectivity matrix (Griffith and Csillag 1993; Griffith and Layne 1997). Hence, behavior of Eq. (14) would be similar even if the distance-based $\mathbf{C}$ is replaced with a binary connectivity-based one.

4.2 Estimation

Equation (14) may be rewritten as

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{E}\mathbf{V}(\boldsymbol{\theta})\mathbf{u} + \boldsymbol{\varepsilon}, \quad \mathbf{u} \sim N(0, \sigma^2\mathbf{I}), \quad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I}), \tag{15}$$

$\mathbf{V}(\boldsymbol{\theta})\mathbf{u} = \boldsymbol{\gamma}$, where $\boldsymbol{\theta} = \{\sigma_\gamma^2/\sigma^2, \alpha\}$, and $\mathbf{V}(\boldsymbol{\theta})$ is a $L \times L$ diagonal matrix whose (l, l)-th element is $(\sigma_\gamma/\sigma)\lambda_l(\alpha)^{1/2}$. Equation (15) is identical to the LMM, whose efficient ML and residual ML (REML) estimations are discussed by Bates (2010) and implemented in "lme4," which is one of the standard packages for mixed effects model estimation in *R* (http://www.r-project.org/). Bates shows that the log-likelihood of Eq. (15) can be expressed as

$$\log lik(\boldsymbol{\beta}, \boldsymbol{\theta}, \sigma^2) = -\frac{N}{2}\log(2\pi\sigma^2) - \log(|\mathbf{V}(\boldsymbol{\theta})|^2) - \frac{d(\boldsymbol{\theta})}{\sigma^2} \tag{16}$$

$$d(\boldsymbol{\theta}) = ||\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{E}\mathbf{V}(\boldsymbol{\theta})\mathbf{u}||^2 + ||\mathbf{u}||^2$$

Note that $||\mathbf{u}||^2$ in $d(\boldsymbol{\theta})$ acts as a regularization term that stabilizes the parameter estimation.

BLUEs of $\boldsymbol{\beta}$ and $\mathbf{u}$ are given by maximizing Eq. (16) as

$$\begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{E}\mathbf{V}(\boldsymbol{\theta}) \\ \mathbf{V}(\boldsymbol{\theta})\mathbf{E}'\mathbf{X} & \mathbf{V}(\boldsymbol{\theta})^2 + \mathbf{I} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{X}'\mathbf{y} \\ \mathbf{V}(\boldsymbol{\theta})\mathbf{E}'\mathbf{y} \end{bmatrix}, \tag{17}$$

where $\mathbf{V}(\boldsymbol{\theta})^2$ is a $L \times L$ diagonal matrix whose (l, l)-th element is $(\sigma_\gamma^2/\sigma^2)\lambda_l(\alpha)$. Equation (17) is identical to Eq. (5) with α added. $\boldsymbol{\theta}$ may be estimated by maximizing Eq. (16). However, ML estimates of variance parameters ($\boldsymbol{\theta}$ and σ^2) are known to be biased (Bates 2010). Hence, we apply the REML estimation, which furnishes unbiased variance parameter estimates. The residual log-likelihood for the variance parameters is formulated as

$$\log lik_R(\boldsymbol{\theta}, \sigma^2) = \int \log lik(\boldsymbol{\beta}, \boldsymbol{\theta}, \sigma^2) d\boldsymbol{\beta}. \tag{18}$$

By maximizing Eq. (18) with respect to σ^2, the well-known unbiased estimator is obtained as

$$\sigma^2 = \frac{||\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{E}\mathbf{V}(\boldsymbol{\theta})\mathbf{u}||^2}{N - K}. \tag{19}$$

$\boldsymbol{\theta}$ is estimated by maximizing its profile residual log-likelihood, $\log lik_R(\boldsymbol{\theta})$, which is derived by the following steps: (a) substitute Eq. (16) into Eq. (18) and derive the function $\log lik_R(\boldsymbol{\theta}, \sigma^2)$, and (b) substitute Eq. (19) into $\log lik_R(\boldsymbol{\theta}, \sigma^2)$ and derive the function $\log lik_R(\boldsymbol{\theta})$ (see, Bates 2010, for more detail). The resulting profile residual log-likelihood function is given by

$$\log lik_R(\boldsymbol{\theta}) = -\frac{1}{2}\log\left|\begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{E}\mathbf{V}(\boldsymbol{\theta}) \\ \mathbf{V}(\boldsymbol{\theta})\mathbf{E}'\mathbf{X} & \mathbf{V}(\boldsymbol{\theta})^2 + \mathbf{I} \end{bmatrix}\right| - \frac{N-K}{2}\left(1 + \log\left(\frac{2\pi d(\boldsymbol{\theta})}{N-K}\right)\right). \tag{20}$$

Maximization of Eq. (20) requires calculating the determinant in the first term. However, the computational complexity, $O((K + L)^3)$, is usually smaller than $O(N^3)$, which is required for ML or REML estimation of the standard geostatistical model (see, Cressie and Wikle 2011). Besides, the computational burden can be lightened by reducing the number of eigenvectors in $\mathbf{E}$. Fortunately, Hughes and Haran (2013) demonstrate, based on their simulation results, that 50–100 eigenvectors may suffice for inferences for $\boldsymbol{\beta}$ in most analysis with RE-ESF (without α).

The parameter estimation procedure of our extended RE-ESF, Eq. (14), may be summarized as follows: (1) estimate $\boldsymbol{\theta}$ by maximizing Eq. (20), where $\boldsymbol{\beta}$ and $\mathbf{u}$ in $d(\boldsymbol{\theta})$ are given by Eq. (17); (2) estimate $\boldsymbol{\beta}$ and $\boldsymbol{\gamma} = \mathbf{V}(\boldsymbol{\theta})\mathbf{u}$ by substituting $\hat{\theta}$ into Eq. (17); and (3) estimate σ^2 by substituting the estimated parameters into Eq. (19).

Variances of the estimated $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ are given as follows

$$\mathrm{Var}\begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\boldsymbol{\gamma}} \end{bmatrix} = \mathrm{Var}\begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \mathbf{V}(\boldsymbol{\theta})\hat{\mathbf{u}} \end{bmatrix} = \sigma^2 \begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{E} \\ \mathbf{E}'\mathbf{X} & \mathbf{I} + \mathbf{V}(\boldsymbol{\theta})^{-2} \end{bmatrix}^{-1}, \tag{21}$$

where $\mathbf{V}(\boldsymbol{\theta})^{-2}$ is a $L \times L$ diagonal matrix whose l-th element is $(\sigma^2/\sigma_\gamma^2)\lambda_l(\alpha)^{-1}$.

Table 1 Comparison of ESF models

	Objective function	α	$\mathbf{M} = \mathbf{I}–\mathbf{1}\mathbf{1}'/N$	$\mathbf{M} = \mathbf{I}–\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$
ESF	Adj.R^2	NA	E(1)	E(X)
	Residual MC		$E_{MC}(1)$	$E_{MC}(X)$
RE-ESF	Residual log-likelihood	Implicitly fixed by one	RE(1)	RE(X)
		Estimated	$RE_{\alpha}(1)$	$RE_{\alpha}(X)$

E(1) and E(X) are adjusted-R^2 maximization-based standard ESF models, $E_{MC}(1)$ and $E_{MC}(X)$ are residual MC minimization-based standard ESF models, RE(1) and RE(X) are RE-ESF models without estimation of α, and $RE_{\alpha}(1)$ and $RE_{\alpha}(X)$ are RE-ESF models with estimation of α. E(1), $E_{MC}(1)$, RE(1), and $RE_{\alpha}(1)$ assume $\mathbf{M} = \mathbf{I}–\mathbf{1}\mathbf{1}'/N$, and E(X), $E_{MC}(X)$, RE(X), and $RE_{\alpha}(X)$ assume $\mathbf{M} = \mathbf{I}–\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$

5 A simulation study

5.1 Assumption

This section compares the standard liner model (LM) and a suite of ESF models with a simulation study. Table 1 summarizes the ESF models compared. Standard ESF models compared include E(1) and E(X), which perform the adjusted-R^2 maximization-based stepwise eigenvector selection, and $E_{MC}(1)$ and $E_{MC}(X)$, which iterate the stepwise eigenvector selection until residual MC becomes less than 0.01 (see, Tiefelsdorf and Griffith 2007). RE-ESF models compared include RE(1) and RE(X), which do not include α, and $RE_{\alpha}(1)$ and $RE_{\alpha}(X)$, which include α. These models are estimated by REML. E(1), E(X), $E_{MC}(1)$, and $E_{MC}(X)$ are discussed by Griffith (2003), and RE(X) is discussed by Hughes and Haran (2013). However, based on our discussion, $RE_{\alpha}(1)$, which we developed in Sects. 3 and 4, might be preferable.

The simulation compares the effectiveness of these ESF models and LM in terms of parameter estimation and inference. Simulation data are generated by Eq. (22), which describes a Gaussian process (see Sect. 3.1):

$$\mathbf{y} = \beta_0\mathbf{1} + \mathbf{x}_1\beta_1 + \mathbf{x}_2\beta_2 + \mathbf{E}\gamma + \boldsymbol{\varepsilon}, \quad \boldsymbol{\gamma} \sim N(\mathbf{0}, \sigma_\gamma^2 k\boldsymbol{\Lambda}(\alpha)), \quad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I}). \tag{22}$$

where β_0 is the intercept, $\mathbf{x}_1$ and $\mathbf{x}_2$ are vectors of explanatory variables, and β_1 and β_2 are their coefficients. $k = N/\sum_l \lambda_l^\alpha$ is a scaling parameter[7] that forces $\mathrm{Var}[\mathbf{E}\boldsymbol{\gamma}] = \sigma_\gamma^2$. Introduction of k allows a comparison between the variances $\mathrm{Var}[\mathbf{E}\boldsymbol{\gamma}]$ and $\mathrm{Var}[\boldsymbol{\varepsilon}]$ using σ_γ^2 and σ^2, respectively. We assume a moderate sample size of 400. Following Seya et al. (2014a), spatial coordinates (x, y) of the simulation data are generated from a standard normal distribution.

Unlike Hughes and Haran (2013), we assume $\mathbf{M} = \mathbf{I}–\mathbf{1}\mathbf{1}'/N$, because the correlation between explanatory variables and $\mathbf{E}\boldsymbol{\gamma}$, which is eliminated when

[7] Expanding the variance of $\mathbf{E}\boldsymbol{\gamma}$ in Eq. (22) as $\mathrm{Var}[\mathbf{E}\gamma] = \frac{\gamma'\mathbf{E}'\mathbf{E}\gamma}{N} = \frac{\gamma'\gamma}{N} = \frac{1}{N}\sum_l \mathrm{diag}[\sigma_\gamma^2 k\boldsymbol{\Lambda}(\alpha)]_l = k\frac{\sigma_\gamma^2}{N}\sum_l \lambda_l^\alpha$, where $\mathrm{diag}[\cdot]_l$ returns the lth diagonal of the matrix $\cdot$. This equation shows that $\mathrm{Var}[\mathbf{E}\boldsymbol{\gamma}]$ equals σ_γ^2 when $k = \frac{N}{\sum_l \lambda_l^\alpha}$.

$\mathbf{M} = \mathbf{I}–\mathbf{X}(\mathbf{X}'\ \mathbf{X})^{-1}\mathbf{X}'$ is used, is possible in most real-world cases. Following Hughes and Haran (2013), eigenvectors corresponding to positive eigenvalues are introduced into $\mathbf{E}$ in RE-ESF models. In standard ESF models, these eigenvectors are assumed to be candidates to be entered into $\mathbf{E}$ in their stepwise eigenvector selection.

One purpose of this simulation was to assess the impact of a spatial confounder on parameter estimates and inferences, i.e., to what extent the correlation between explanatory variables and $\mathbf{E}\boldsymbol{\gamma}$ makes parameter estimates unstable. To examine it, we generate $\mathbf{x}_1$ and $\mathbf{x}_2$ using Eq. (23) and Eq. (24), respectively:

$$\mathbf{x}_1 = \mathbf{E}\gamma_{x1} + \boldsymbol{\varepsilon}_{x1}, \quad \boldsymbol{\gamma}_{x1} \sim N(\mathbf{0}, \sigma^2_{\gamma(x)} k\boldsymbol{\Lambda}(\alpha)), \quad \boldsymbol{\varepsilon}_{x1} \sim N(\mathbf{0}, (1-\sigma_{\gamma(x)})^2\mathbf{I}), \tag{23}$$

$$\mathbf{x}_2 = \mathbf{E}\gamma_{x2} + \boldsymbol{\varepsilon}_{x2}, \quad \boldsymbol{\gamma}_{x1} \sim N(\mathbf{0}, \sigma^2_{\gamma(x)} k\boldsymbol{\Lambda}(\alpha)), \quad \boldsymbol{\varepsilon}_{x1} \sim N(\mathbf{0}, (1-\sigma_{\gamma(x)})^2\mathbf{I}). \tag{24}$$

where $\sigma^2_{\gamma(x)}$ is the rate of spatial-dependent variation accounting for the total variation in $\mathbf{x}_1$ or $\mathbf{x}_2$. When $\sigma^2_{\gamma(x)} = 0$, $\mathbf{x}_1$ and $\mathbf{x}_2$ do not have any spatial-dependent component, and accordingly, the correlation between these explanatory variables and $\mathbf{E}\boldsymbol{\gamma}$ is approximately zero (i.e., no spatial confounding). In contrast, when $\sigma^2_{\gamma(x)} \neq 0$, $\mathbf{x}_1$ and $\mathbf{x}_2$ have a spatial dependence component, and they are correlated with $\mathbf{E}\boldsymbol{\gamma}$.

True parameter values are set as follows: $\alpha = 4.0$, $\beta_1 = 1.0$, and $\beta_2 = -0.4$. Their estimates and standard errors are evaluated by varying variance parameter values: $\sigma^2 = 1.0$, $\sigma^2_\gamma \in \{0.0, 1.0, 2.0^2\}$, $\alpha \in \{1, 3, 5\}$, and $\sigma^2_{\gamma(x)} \in \{0, 0.4^2, 0.8^2\}$. Each case has 2000 replications. RE(1) and RE(X) implicitly assume $\alpha = 1$.

All our calculations are implemented on a Windows 7 64-bit system with a 48 GB memory, and coded using R (version 3.1.1).

6 Results

The accuracy of estimated trend parameters, β_1 and β_2, is evaluated by the root mean squared error (RMSE), which is given by

$$\mathrm{RMSE}(\beta_p) = \sqrt{\frac{1}{2000}\sum\nolimits_{\mathrm{iter}=1}^{2000} (\widehat{\beta}_p^{(\mathrm{iter})} - \beta_p)^2}, \tag{25}$$

where iter denotes iteration number, $\hat{\beta}_p^{(\mathrm{iter})}$ is β_p estimated in the *iter*-th iteration, and $p \in \{1, 2\}$. Figure 2 summarizes the RMSEs of β_1 and β_2 estimated by LM, E(1), $\mathrm{E_{MC}}(1)$, RE(1), and $\mathrm{RE}_\alpha(1)$. Note that estimates of E(X), $\mathrm{E_{MC}}(\mathrm{X})$, RE(X), and $\mathrm{RE}_\alpha(\mathrm{X})$ are always equal to those of LM.[8]

[8] Equation (5) is expanded using $\mathbf{X}'\ \mathbf{E} = \mathbf{0}$, which $\mathbf{M} = \mathbf{I}–\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ implies, as $\begin{bmatrix}\hat{\beta}\\ \hat{\gamma}\end{bmatrix} = \begin{bmatrix}\mathbf{X}'\mathbf{X} & \mathbf{0}\\ \mathbf{0} & \mathbf{I}+\frac{\sigma^2}{\sigma^2_\gamma}\boldsymbol{\Lambda}^{-1}\end{bmatrix}^{-1}\begin{bmatrix}\mathbf{X}'\mathbf{y}\\ \mathbf{E}'\mathbf{y}\end{bmatrix} = \begin{bmatrix}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}\\ \left(\mathbf{I}+\frac{\sigma^2}{\sigma^2_\gamma}\boldsymbol{\Lambda}^{-1}\right)^{-1}\mathbf{E}'\mathbf{y}\end{bmatrix}$. Thus, $\hat{\beta}$ equals the estimate of the OLS estimate.

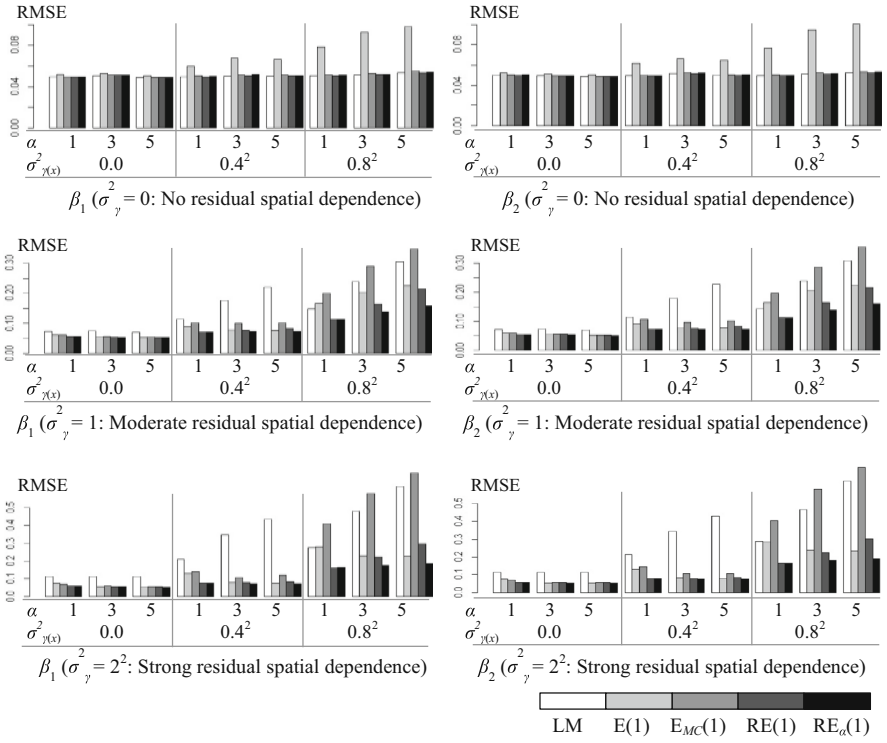

Fig. 2 RMSEs of estimated β_1 (*left*) and β_2 (*right*): The *top* ($\sigma^2_\gamma = 0$), the *middle* ($\sigma^2_\gamma = 1$), and the *bottom* ($\sigma^2_\gamma = 2$) part of the figure corresponds to cases with no, moderate, and strong residual spatial dependences, respectively. In each case, RMSEs are evaluated by varying $\sigma^2_{\gamma(x)}$, which describes the rate of spatial-dependent variation in explanatory variables, and α, which describes the spatial scale of the spatial-dependent variations

E(1) indicates larger RMSEs when residuals have no spatial dependence ($\sigma^2_\gamma = 0$). To examine the reason, average numbers of selected eigenvectors are summarized in Fig. 3. This figure suggests that E(1) selects around 12 eigenvectors even in cases with an absence of spatial dependence. The excessively selected eigenvectors would be the source of large RMSEs. In contrast, $E_{MC}(1)$ selects less than one eigenvector on average and indicates smaller RMSEs. RE(1) and $RE_\alpha(1)$ also indicate small RMSEs. Thus, in the absence of residual spatial dependence, $E_{MC}(1)$, RE(1), and $RE_\alpha(1)$ provide accurate parameter estimates.

However, the RMSEs of $E_{MC}(1)$ inflate when both residuals and explanatory variables are spatially dependent ($\sigma^2_\gamma \neq 0$ and $\sigma^2_{\gamma(x)} \neq 0$), while RE(1) and $RE_\alpha(1)$ show a relatively smaller inflation of RMSEs even in this case. Besides, $RE_\alpha(1)$ displays smaller RMSEs than RE(1) in many cases. In particular, the gap between $RE_\alpha(1)$ and RE(1) is significant when residuals and explanatory variables have strong spatial dependence, and α is far from one, which RE(1) implicitly assumes. RMSEs of our proposed $RE_\alpha(1)$ are fairly small across cases.

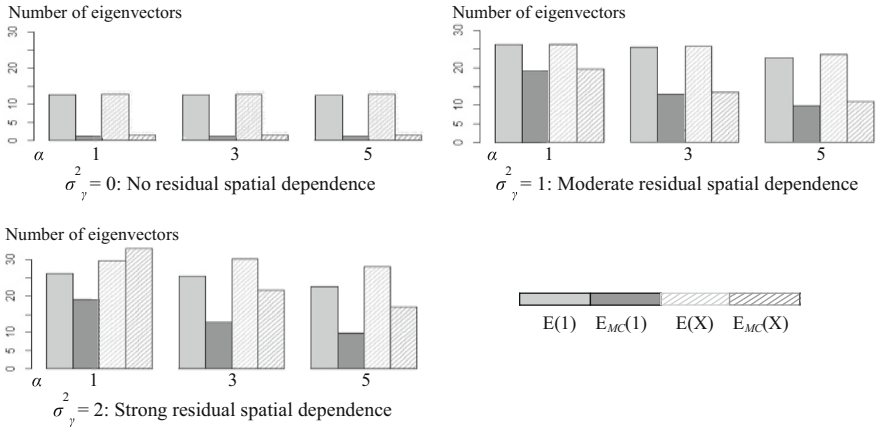

Fig. 3 Number of selected eigenvectors (E(1), $E_{MC}(1)$, E(X), $E_{MC}(X)$): *top left* (no spatial dependence: $\sigma_\gamma^2 = 0$); *top right* (moderate spatial dependence: $\sigma_\gamma^2 = 1$); *bottom left* (strong spatial dependence: $\sigma_\gamma^2 = 2$)

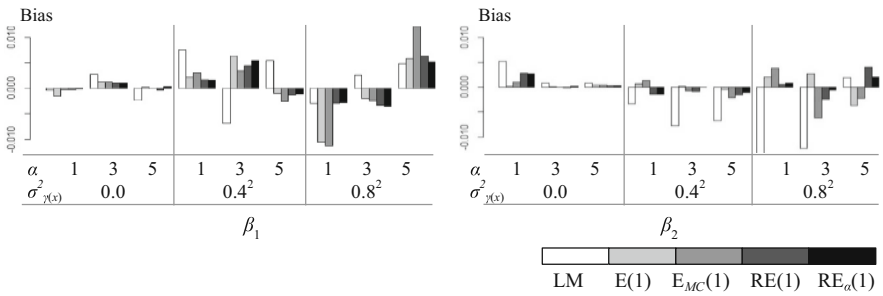

Fig. 4 Biases of estimated β_1 (*left*) and β_2 (*right*): here, biases in strong residual spatial dependence (i.e., $\sigma_\gamma^2 = 2^2$) cases are displayed

Another interesting result is that, when residuals are spatially dependent, RMSEs of E(1) are comparable with RE(1) and $RE_\alpha(1)$. This result demonstrates that the standard ESF provides accurate parameter estimates. In contrast, large RMSEs of LM suggest inefficiency of models with $\mathbf{M} = \mathbf{I}–\mathbf{X}(\mathbf{X}'\,\mathbf{X})^{-1}\mathbf{X}'$, whose estimates are identical to those for LM. This inefficiency would be due to their strong assumption that the spatial dependence component in residuals is uncorrelated with $\mathbf{X}$ (Hodges and Reich 2010).Next, biases of estimated parameters are evaluated using Eq. (26):

$$\text{Bias}(\beta_p) = \frac{1}{2000}\sum_{\text{iter}=1}^{2000} (\hat{\beta}_p^{(\text{iter})} - \beta_p). \tag{26}$$

In each model, biases are at most $\pm$ 0.01, which is sufficiently small relative to their true values. For reference, biases when $\sigma_\gamma^2 = 2^2$, in which biases are largest, are shown in Fig. 4.

To compare the effectiveness of the models for statistical inferences, t-values of $\hat{\beta}_p$ are compared to their true values. The estimated t-value in the *iter*-th iteration is calculated by $\hat{t}_p^{(\mathrm{iter})} = \hat{\beta}_p / \sqrt{\mathrm{Var}(\hat{\beta}_p)^{(\mathrm{iter})}}$, where $\mathrm{Var}(\hat{\beta}_p)^{(\mathrm{iter})}$ is the variance of $\hat{\beta}_p$, which is calculated by substituting estimated parameters into Eq. (21), and the true t-value, $t_p^{(\mathrm{iter})} = \beta_p / \sqrt{\mathrm{Var}(\beta_p)^{(\mathrm{iter})}}$, is evaluated by substituting true parameter values into Eq. (21). Note that, as a result of changing $\mathbf{x}_1$ and $\mathbf{x}_2$ across iterations, true t-values in each iteration are not constant. To compare the accuracy of t-value estimates across cases whose true values differ from each other, the root mean squared percentage error (RMSPE), which is defined as follows, is applied:

$$\mathrm{RMSPE}(t_p) = \sqrt{\frac{1}{2,000} \sum_{\mathrm{iter}=1}^{2,000} \left(\frac{\hat{t}_p^{(\mathrm{iter})} - t_p^{(\mathrm{iter})}}{t_p^{(\mathrm{iter})}} \right)^2}. \tag{27}$$

Figures 5 and 6 summarize RMSPEs of t_1 and t_2, respectively. As a whole, RMSPEs are large when explanatory variables have spatial dependence ($\sigma^2_{\gamma(x)} \neq 0$).

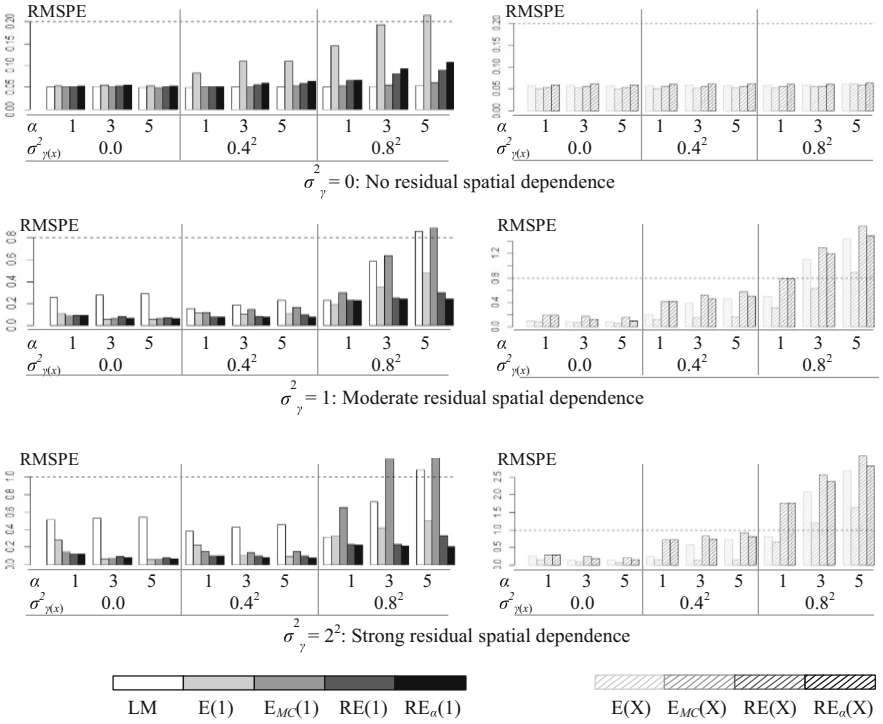

Fig. 5 RMSPEs of t-values (β_1): The *left* figures summarize RMSPEs of LM and ESF models with $\mathbf{M} = \mathbf{I}–\mathbf{1}\mathbf{1}'/N$, and the *right* figures summarize RMSPE values of ESF models with $\mathbf{M} = \mathbf{I}–\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$. The *top* ($\sigma^2_\gamma = 0$), the *middle* ($\sigma^2_\gamma = 1$), and the *bottom* ($\sigma^2_\gamma = 2$) part of the figure corresponds to cases with no, moderate, and strong residual spatial dependences, respectively. *Dash lines* are introduced to compare left and right figures, whose scales are different

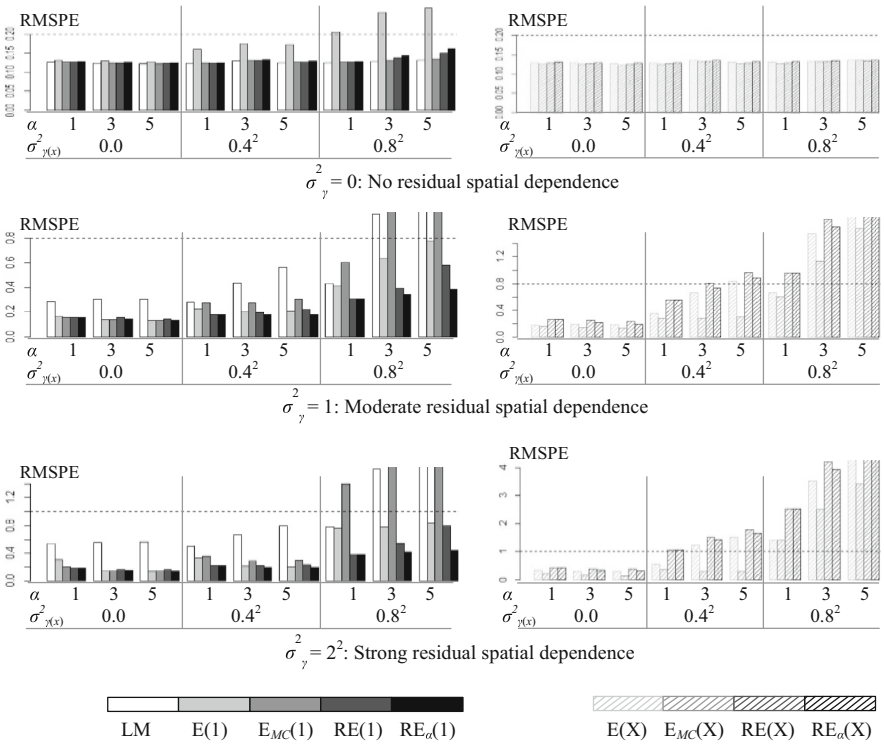

Fig. 6 RMSPEs of *t*-values (β_2): The left figures summarize RMSPEs of LM and ESF models with $\mathbf{M} = \mathbf{I}–\mathbf{11}'/N$, and the right figures summarize RMSPE values of ESF models with $\mathbf{M} = \mathbf{I}–\mathbf{X}(\mathbf{X}'\ \mathbf{X})^{-1}\mathbf{X}'$. The *top* ($\sigma^2_\gamma = 0$), the *middle* ($\sigma^2_\gamma = 1$), and the *bottom* ($\sigma^2_\gamma = 2$) part of the figure corresponds to cases with no, moderate, and strong residual spatial dependences, respectively. *Dash lines* are introduced to compare *left* and *right* figures, whose scales are different

The tendency is remarkable when $\mathbf{M} = \mathbf{I}–\mathbf{X}(\mathbf{X}'\ \mathbf{X})^{-1}\mathbf{X}'$ is used. Thus, inefficiency of $\mathbf{M} = \mathbf{I}–\mathbf{X}(\mathbf{X}'\ \mathbf{X})^{-1}\mathbf{X}'$ is again confirmed. Among models with $\mathbf{M} = \mathbf{I}–\mathbf{11}'/N$, as holds for the RMSEs of β_1 and β_2, E(1) outperforms $E_{MC}(1)$ when residuals are spatially dependent, and otherwise, the reverse is true. RMSPEs of RE(1) and $RE_\alpha(1)$ are smaller than E(1) and $E_{MC}(1)$ when explanatory variables are spatially dependent ($\sigma^2_{\gamma(x)} \neq 0$). Furthermore, RMSPEs of $RE_\alpha(1)$ are smaller than those for RE(1) in many cases, in particular, when residual and explanatory variables have strong spatial dependence, and α is far from one, which RE(1) assumes. The difference between RE(1) and $RE_\alpha(1)$ is especially large in t_2, whose true coefficient β_2 is small. Thus, the effectiveness of $RE_\alpha(1)$ again is verified in terms of accuracy of the *t*-value estimates. Although the RMSEs of E(1) are larger than those for the RE-ESF models, these values are comparable as long as spatial dependence in $\mathbf{X}$ is not significant ($\sigma^2_{\gamma(x)} = 0.8$). Considering the simplicity of E(1), it would still be an effective alternative.

Figure 7 portrays biases of *t*-values. This figure shows that *t*-values are seriously overestimated when $\mathbf{M} = \mathbf{I}–\mathbf{X}(\mathbf{X}'\ \mathbf{X})^{-1}\mathbf{X}'$ is assumed. Among the other models, *t*-

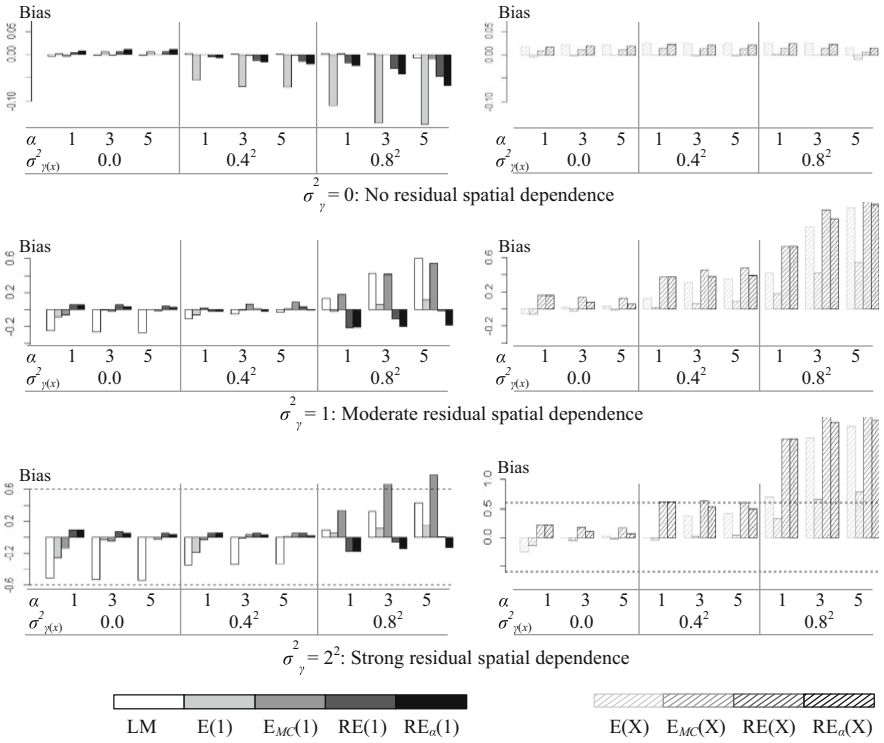

Fig. 7 Biases of t-values (β_2). The left figures summarize biases of LM and ESF models with $\mathbf{M} = \mathbf{I}$–$\mathbf{1}\mathbf{1}'/N$, and the right figures summarize biases of ESF models with $\mathbf{M} = \mathbf{I}$–$\mathbf{X}(\mathbf{X}'\ \mathbf{X})^{-1}\mathbf{X}'$. The *top* ($\sigma_\gamma^2 = 0$), the *middle* ($\sigma_\gamma^2 = 1$), and the *bottom* ($\sigma_\gamma^2 = 2$) part of the figure corresponds to cases with no, moderate, and strong residual spatial dependences, respectively. *Dash lines* are introduced to compare left and right figures, whose scales are different

values of E(1) tend to be underestimated when residual spatial dependence is absent, while t-values of LM and $E_{MC}(1)$ tend to be upwardly biased in cases with a presence of residual spatial dependence. Although biases in RE(1) and $RE_\alpha(1)$ are relatively small, the latter tends to have greater bias than the former. This might be due to the overestimation of α, which we subsequently discuss.

Figure 8 describes the RMSEs and biases of α in $RE_\alpha(1)$. These RMSEs and biases are sufficiently small compared to the true value of α. They demonstrate that our REML provides fairly accurate estimates of α. Still, RMSEs increase when the true value of α is large and residual spatial dependence variation is small ($\sigma_\gamma^2 = 1$). Besides, α tends to be overestimated in these cases. Reducing RMSEs and biases would be important to enhance the efficiency of $RE_\alpha(1)$ for trend parameter estimation inferences.

Finally, Fig. 9 summarizes average computational times. Except for LM, RE(1) is the fastest (average: 0.65 s.), and our proposed $E_\alpha(1)$ is also fast (average: 1.02 s.) compared to the standard ESF-type models, which must perform a stepwise eigenvector selection. This result would be useful to extend RE-ESF models to large

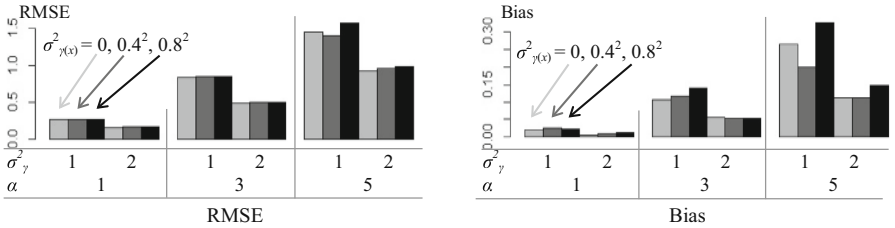

Fig. 8 RMSEs (*left*) and biases (*right*) of α ($RE_\alpha(1)$)

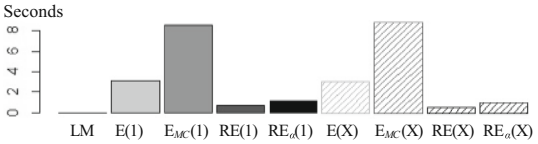

Fig. 9 Average computation time: LM is the standard linear regression model. E(1) and E(X) are adjusted-R^2 maximization-based standard ESF models, $E_{MC}(1)$ and $E_{MC}(X)$ are the residual MC minimization-based ESF models, RE(1) and RE(X) are RE-ESF models without estimation of α, and $RE_\alpha(1)$ and $RE_\alpha(X)$ are RE-ESF models with estimation of α. E(1), $E_{MC}(1)$, RE(1), and $RE_\alpha(1)$ assume $\mathbf{M} = \mathbf{I}–\mathbf{11}'/N$, and E(X), $E_{MC}(X)$, RE(X), and $RE_\alpha(X)$ assume $\mathbf{M} = \mathbf{I}–\mathbf{X}(\mathbf{X}'\ \mathbf{X})^{-1}\mathbf{X}'$

datasets. Actually, the residual log-likelihood of our model can be maximized efficiently even for large data, provided that less than 100 eigenvectors are considered, as suggested by Hughes and Haran (2013) (i.e., 50 to 100 eigenvectors may suffice for RE-ESF-based statistical inferences). Although eigendecomposition is computationally intensive, it can be approximated efficiently, e.g., by applying the ARPACK procedure, which is applied for ESF by Seya et al. (2014b), the Nyström extension (Drineas and Mahoney 2005), and an analytic approximation for regular square tessellations, which is proposed by Griffith (2000).

In summary, our developed $RE_\alpha(1)$ provides accurate parameter estimates and *t*-values. These accuracies are stable across cases compared to the other models. We demonstrate the computational efficiency of $RE_\alpha(1)$ and also suggest that the use of $\mathbf{M} = \mathbf{I}–\mathbf{X}(\mathbf{X}'\ \mathbf{X})^{-1}\mathbf{X}'$ can lead to misleading results, and that the effectiveness of E(1) is comparable in many cases.

7 Concluding remarks

This study first discusses the theoretical properties of conventional ESF models and a RE-ESF version by Hughes and Haran (2013). There, we show that, although the RE-ESF model is theoretically consistent with the standard geostatistical model, it can be sensitive to the chosen spatial connectivity matrix, **C**. To mitigate the dependency on **C**, we extend the RE-ESF to estimate the scale of a spatial dependency structure endogenously (see Fig. 1) and introduce a computationally efficient REML estimation for the model.

A Monte Carlo simulation shows the effectiveness of our RE-ESF, $RE_{\alpha}(1)$; it provides accurate parameter estimates across cases, whereas the other models provide less accurate estimates in some cases; especially, models with $\mathbf{M} = \mathbf{I}–\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ indicate erroneous results in many cases. This is due to the strong assumption that residual spatial components are uncorrelated with explanatory variables (Hodges and Reich 2010). Development of another type of $\mathbf{M}$ that can cope with the spatial confounding bias (just like $\mathbf{M} = \mathbf{I}–\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$), while considering the spatial component correlated with $\mathbf{X}$ (just like $\mathbf{M} = \mathbf{I}–\mathbf{1}\mathbf{1}'/N$), would be an important future research endeavor.

The stability of $RE_{\alpha}(1)$ would be a good reason to extend it, e.g., non-Gaussian models and spatially varying coefficient models (see, Griffith 2008). The stability might hold even if data have some missing values (see, Griffith 2004b). Further, $RE_{\alpha}(1)$, which does not require stepwise eigenvector selection, also might be useful for large data modeling, provided that eigenvectors of a large spatial weights matrix $\mathbf{C}$ can be approximated efficiently, for example, using the ARPACK procedure, the Nyström extension, and an analytic approximation proposed by Griffith (2000). These extensions would be interesting topics that advance discussion of ESF models.

R code for the proposed method is available on request (please e-mail murakami.daisuke@nies.go.jp or murakami_daisuke_t@yahoo.co.jp).

Acknowledgments We received helpful comments from Professor Arthur Getis at the 59th Annual North American Meetings of the Regional Science Association International. Also, we received generous support from Professor Morito Tsutsumi, University of Tsukuba, and Professor Yoshiki Yamagata, National Institute for Environmental Studies. We would particularly like to thank them. In addition, we are grateful to anonymous referees for their many suggestive comments. This work was supported by a grant-in-aid from the Japan Society for the Promotion of Science Fellows. In addition, this is an achievement of the Global Climate Risk Management Strategies (S10) project.

References

Anselin L, Rey S (1991) Properties of tests for spatial dependence in linear regression models. Geogr Anal 23(2):112–131

Arbia G, Fingleton B (2008) New spatial econometric techniques and applications in Regional Science. Pap Reg Sci 87(3):311–317

Bates DM (2010) lme4: Mixed-effects modeling with R. http://lme4.r-forge.r-project.org/book

Blancheta FG, Legendre P, Borcard D (2008) Modelling directional spatial process in ecological data. Ecol Model 215(4):325–336

Borcard D, Legendre P (2002) All-scale spatial analysis of ecological data by means of principal coordinates of neighbor matrices. Ecol Model 153(1–2):51–68

Chun Y (2008) Modeling network autocorrelation within migration flows by eigenvector spatial filtering. J Geogr Syst 10(4):317–344

Chun Y (2014) Analyzing space–time crime incidents using eigenvector spatial filtering: an application to vehicle burglary. Geogr Anal 46(2):165–184

Chun Y, Griffith DA (2011) Modeling network autocorrelation in space–time migration flow data: an eigenvector spatial filtering approach. Ann Assoc Am Geogr 101(3):523–536

Chun Y, Griffith DA (2013) Spatial statistics and geostatistics: theory and applications for geographic information science and technology. SAGE, London

Chun Y, Griffith DA (2014) A quality assessment of eigenvector spatial filtering based parameter estimates for the normal probability model. Spat Stat 10:1–11

Cressie N (1993) Statistics for spatial data. John Wiley & Sons, New York

Cressie N, Wikle CK (2011) Statistics for spatio-temporal data. Wiley, New York
Cuaresma CJ, Feldkircher M (2013) Spatial filtering, model uncertainty and the speed of income convergence in Europe. J Appl Econom 28(4):720–741
Dray S, Legendre P, Peres-Neto PR (2006) Spatial modeling: a comprehensive framework for principal coordinate analysis of neighbor matrices (PCNM). Ecol Model 196(3–4):483–493
Drineas P, Mahoney MW (2005) On the Nyström method for approximating a gram matrix for improved kernel-based learning. J Mach Learn Res 6:2153–2175
Elhorst JP (2010) Applied spatial econometrics: raising the bar. Spat Econ Anal 5(1):9–28
Fischer MM, Griffith DA (2008) Modeling spatial autocorrelation in spatial interaction data: an application to patent citation data in European Union. J Reg Sci 48(5):969–989
Franzese RJ, Hays JC (2014) Testing for spatial-autoregressive lag versus (unobserved) spatially correlated error-components. Benjamin F. Shambaugh conference: New frontiers in the study of policy diffusion, University of Iowa, Iowa City
Getis A (1990) Screening for spatial dependence in regression analysis. Paper Reg Sci Assoc 69(1):69–81
Getis A, Griffith DA (2002) Comparative spatial filtering in regression analysis. Geogr Anal 34(2):130–140
Getis A, Ord K (1992) The analysis of spatial association by use of distance statistics. Geogr Anal 24(3):189–206
Griffith DA (1996) Spatial autocorrelation and eigenfunctions of the geographic weights matrix accompanying georeferenced data. Can Geogr 40(4):351–367
Griffith DA (2000) Eigenfunction properties and approximations of selected incidence matrices employed in spatial analyses. Linear Algebr Appl 321(1–3):95–112
Griffith DA (2002) A spatial filtering specification of the auto-Poisson model. Stat Prob Letters 58(3):245–251
Griffith DA (2003) Spatial autocorrelation and spatial filtering: Gaining understanding through theory and scientific visualization. Springer, Berlin
Griffith DA (2004a) A spatial filtering specification for the autologistic model. Environ Plan 36(10):1791–1811
Griffith DA (2004b) Distributional properties of georeferenced random variables based on the eigenfunction spatial filter. J Geogr Syst 6(3):263–288
Griffith DA (2006) Hidden negative spatial autocorrelation. J Geogr Syst 8(4):335–355
Griffith DA (2008) Spatial-filtering-based contributions to a critique of geographically weighted regression (GWR). Environ Plan 40(11):2751–2769
Griffith DA (2009) Modeling spatial autocorrelation in spatial interaction data: empirical evidence from 2002 Germany journey-to-work flows. J Geogr Syst 11(2):117–140
Griffith DA (2010) Spatial filtering. In: Fischer MM, Getis A (eds) Handbook of applied spatial analysis. Springer, Berlin, pp 301–318
Griffith DA (2011) Visualizing analytical spatial autocorrelation components latent in spatial interaction data: an eigenvector spatial filter approach. Comput Environ Urban Syst 35(2):140–149
Griffith DA, Chun Y (2013) Spatial autocorrelation and spatial filtering. In: Fischer MM, Nijkamp P (eds) Handbook of regional science. Springer, Berlin, pp 1477–1507
Griffith DA, Csillag F (1993) Exploring relationship between semi-variogram and spatial autoregressive models. Paper Reg Sci 72(3):283–295
Griffith DA, Fischer MM (2013) Constrained variants of the gravity model and spatial dependence: model specification and estimation issues. J Geogr Syst 15(3):291–317
Griffith DA, Layne L (1997) Uncovering relationships between geo-statistical and spatial autoregressive models. In: American Statistical Association (ed) Proceedings of the section on statistics and the environment 1997, American Statistical Association, Alexandria, Virginia, pp 91–96
Griffith DA, Paelinck JHP (2011) Non–standard spatial statistics and spatial econometrics. Springer, Berlin
Griffith DA, Peres-Neto PR (2006) Spatial modeling in ecology: the flexibility of eigenfunction spatial analyses in exploiting relative location information. Ecol 87(10):2603–2613
Grimpe C, Patuelli R (2009) Regional knowledge production in nanomaterials: a spatial filtering approach. Ann Reg Sci 46(3):519–541
Haskard KA, Lark RM (2009) Modelling non-stationary variance of soil properties by tempering an empirical spectrum. Geoderma 153(1–2):18–28
Helbich M, Arsanjani JJ (2014) Spatial eigenvector filtering for spatiotemporal crime mapping and spatial crime analysis. Cartogr Geogr Inf Sci 42(2):134–148

Henderson CR (1975) Best linear unbiased estimation and prediction under a selection model. Biometrics 31(2):423–447

Hodges JS (2013) Richly parameterized linear models: additive, time series, and spatial models using random effects. CRC Press, Boca Raton

Hodges JS, Reich BJ (2010) Adding spatially-correlated errors can mess up the fixed effect you love. Am Stat 64(4):325–334

Hughes J, Haran M (2013) Dimension reduction and alleviation of confounding for spatial generalized linear mixed models. J R Stat Soc B Stat Methodol 75(1):139–159

Jacob GB, Muturi EJ, Caamano EX, Gunter JT et al (2008) Hydrological modeling of geophysical parameters of arboviral and protozoan disease vectors in Internally Displaced People camps in Gulu, Uganda. Int J Health Geogr 7:11

Legendre P, Legendre L (2012) Numerical Ecology, 3rd edn. Elsevier Science BV, Amsterdam

LeSage JP, Pace RK (2009) Introduction to spatial econometrics. Chapman & Hall/CRC, Boca Raton

LeSage JP, Pace RK (2014) The biggest myth in spatial econometrics. Econom 2(4):217–249

LeSage JP, Parent O (2007) Bayesian model averaging for spatial econometric models. Geogr Anal 39(3):241–267

Moniruzzaman M, Paez A (2012) A model-based approach to select case sites for walkability audits. Health Place 18(6):1323–1334

Moran PAP (1950) Notes on continuous stochastic phenomena. Biom 37(1/2):17–23

Pace RK, LeSage JP (2010) Omitted variable biases of OLS and spatial lag models. In: Páez A, Gallo J, Buliung RN, Dall'erba S (eds) Progress in spatial analysis. Springer, Berlin, pp 17–28

Pace RK, LeSage JP, Zhu S (2013) Interpretation and computation of estimates from regression models using spatial filtering. Spat Econ Anal 8(3):352–369

Paciorek CJ (2010) The importance of scale for spatial-confounding bias and precision of spatial regression estimators. Stat Sci 25(1):107–125

Patuelli R, Griffith DA, Tiefelsdorf M, Nijkamp P (2011) Spatial filtering and eigenvector stability: space-time models for German unemployment data. Int Reg Sci Rev 34(2):253–280

Peres-Neto PR, Legendre P, Dray S, Borcard D (2006) Variation partitioning of species data matrices: estimation and comparison of fractions. Ecol Soc Am 87(10):2614–2625

Pintore A, Holmes CC (2004) Non-stationary covariance functions via spatially adaptive spectra. University of Oxford, Oxford

Scherngell T, Lata R (2013) Towards an integrated European Research Area? Findings from eigenvector spatially filtered spatial interaction models using European Framework Programme data. Pap Reg Sci 92(3):555–577

Seya H, Tsutsumi M, Yamagata Y (2014a) Weighted-average least squares applied to spatial econometric models: a Monte Carlo investigation. Geogr Anal 46(2):126–147

Seya H, Murakami D, Tsutsumi M, Yamagata Y (2014b) Application of LASSO to the eigenvector selection problem in eigenvector-based spatial filtering. Geogr Anal. doi:10.1111/gean.12054

Thayn JB, Simanis JM (2013) Accounting for spatial autocorrelation in linear regression models using spatial filtering with eigenvectors. Ann Assoc Am Geogr 103(1):47–66

Tiefelsdorf M, Griffith DA (2007) Semiparametric filtering of spatial autocorrelation: the eigenvector approach. Environ Plan A 39(5):1193–1221

Wang Y, Kockelman KM, Wang X (2013) Understanding spatial filtering for analysis of land use-transport data. J Transp Geogr 31:123–131